

Pharma Commercial Decision Analytics

Synthetic portfolio case study for a Decision Analytics Associate application

Executive objective

Evaluate where and how a fictional pharmaceutical company should commercialize Drug_A for type-2 diabetes in India using market opportunity, physician segmentation, demand forecasting and pricing scenarios.

Top market opportunities

Region	Treated patients	Growth %	Opportunity score
Karnataka	170,753	16.0	92.5
Maharashtra	177,845	12.0	83.3
Telangana	112,197	18.0	81.5
Tamil Nadu	130,747	11.0	67.3
Gujarat	106,609	9.0	55.0

Analytical interpretation

The composite opportunity framework highlights Karnataka, Maharashtra, Telangana as the leading regions in the synthetic dataset. Physician segmentation then identifies distinct commercial groups for differentiated field-force action. The six-month forecast provides a planning baseline, while pricing scenarios show how revenue changes under different assumptions.

Key modeling assumptions

Forecast uses lagged demand, rolling averages and monthly seasonality. Pricing uses a log-log elasticity relationship. Scenario analysis uses $\pm 15\%$ adoption around the base case. These are portfolio assumptions, not real-world estimates.

Recommended executive workflow

1) Prioritize high-opportunity regions. 2) Focus commercial resources on high-value and engaged physician segments. 3) Use the forecast for supply and field planning. 4) Evaluate price scenarios with finance/market-access stakeholders. 5) Monitor actual-vs-forecast performance and update assumptions.

Important: All data and results are synthetic.